

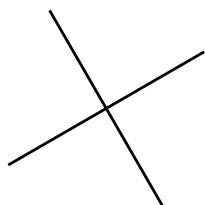
Chemistry — Assignment 2: Degree of Carbon & Hydrogen Atoms

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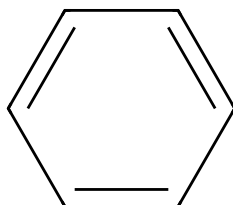
Chemistry (Foundational Chemistry) · Semester 5

Question. Identify the degree of the carbon and hydrogen atoms (count $1^\circ/2^\circ/3^\circ/4^\circ$ C and $1^\circ/2^\circ/3^\circ$ H) for: (1) neopentane, (2) benzene, (3) 9,10-dimethyldecalin (fused bicyclic), (4) cyclohexanone.

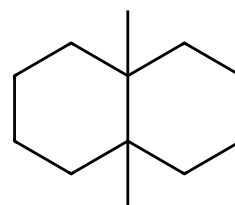
Rule. A carbon is $1^\circ/2^\circ/3^\circ/4^\circ$ by the number of *other carbons* bonded to it. A hydrogen takes the degree of the carbon it sits on. (Counts below computed directly from each structure.)



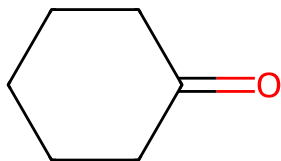
neopentane



benzene



9,10-dimethyldecalin



cyclohexanone

Compound	1°C	2°C	3°C	4°C	1°H	2°H	3°H
neopentane (C ₅ H ₁₂)	4	0	0	1	12	0	0
benzene (C ₆ H ₆)	0	6	0	0	0	6	0
9,10-dimethyldecalin (C ₁₂ H ₂₂)	2	8	0	2	6	16	0
cyclohexanone (C ₆ H ₁₀ O)	0	6	0	0	0	10	0